

BLESSED N. KUTYAUROIPO

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EDUCATION

Mississippi Valley State University (GPA: 4.0/4.0)

Bachelor of Science Engineering Technology (Electronics)

Minor in Computer Science

Awards: Presidential Scholar, Honors Student, FIRST Global Robotics Captain, XPRIZE Award

Itta Bena, MS

Expected May 2028

TECHNICAL SKILLS

Programming: Python, C++, C++17, Java, ROS

Embedded Systems & Electronics: Circuit Design, PCB Prototyping, Soldering, Raspberry Pi, 3D printing

Engineering & Development Tools: Git, GitHub, Google Colab, Figma, UI/UX, LaTeX

AI & Data: YOLOv8, Open CV, XGBoost, LSTMs, CNNs, Pandas

INDEPENDENT ENGINEERING PROJECTS

Occupancy-Aware HVAC Control System

- Designed electrical schematic and embedded hardware architecture integrating Raspberry Pi, magnetic door sensor, DS18B20 temperature sensor, and relay control
- Programmed Raspberry Pi GPIO to acquire real-time sensor inputs and automate HVAC operation based on occupancy and temperature conditions.
- Integrated hardware and software components into a real-time sensing and control system focused on energy-efficient operation.
- Engineered an occupancy-aware HVAC system to improve building energy efficiency while maintaining occupant comfort.

The Electric Step: Design and Evaluation of a Footstep-Based Energy Harvesting System

- Designed and built a piezoelectric energy-harvesting prototype that converts footsteps into electrical energy.
- Integrated an ESP32 microcontroller for real-time monitoring of electrical output.
- Designed the embedded hardware architecture and wireless monitoring system for real-time sensor data acquisition.
- Fabricated and assembled the prototype using 3D-printed structural components.

EXPERIENCE

Team Lead

January 2022 – October 2022

Zimbabwe Robotics

Harare, Zimbabwe

- Led a 6-member robotics team in mechanical design and assembly, integrating REV Smart Robot Servos, Core/HD Hex motors, bearings, gears, and structural components into a competition-ready robot.
- Coordinated mechanical, electrical, and programming efforts across 5+ build/testing milestones, integrating motors and control electronics to ensure reliable robot performance under competition conditions.
- Featured in a **CNN documentary** covering Team Zimbabwe's participation in the FIRST Global Challenge, highlighting our engineering work and international robotics competition experience.

Intern

May 2025 – June 2025

STEM and Enterprise LAUNCH Program

Berlin, Germany

- Led a 6-member team in coordinating project tasks and development milestones, ensuring efficient collaboration and timely completion of assigned STEM project objectives.
- Designed a workflow for a STEM innovation project, supporting solution development and validation during the program.
- Delivered technical presentations of project results to faculty and industry professionals, translating complex findings into clear insights for diverse stakeholders.

Research Intern (AI/ML)

May 2026 – July 2026

Center for Equitable AI and Machine Learning Systems

Baltimore, MD

- Collaborated with a 4-member AI research team to predict PM2.5 using satellite-derived (AOD) and meteorological data.
- Processed, merged, resampled and interpolated thousands of satellite and meteorological observations using Python (Pandas) to prepare analysis-ready datasets.
- Designed, trained, and evaluated several machine learning and deep learning models on satellite and meteorological datasets, assessing predictive performance through RMSE, R^2 , and interactive data visualizations.

LEADERSHIP & PROFESSIONAL INVOLVEMENT

Junior Class Secretary (SGA), Strada Scholar (MVSU), NSBE MVSU Chapter Member, Men's Tennis Team (MVSU), Fellowship of Christian Athletes (FCA), Resident Advisor